Chapter 16. Logistic Regression

Logistic model, odds ratios, maximum likelihood and IRLS, 2×2 tables

Contents

16.1 A model for a binary response **2**

16.2 Estimation **2**

16.3 Two-by-two tables **3**

16.4 Selected exercises **4**

16.1 A model for a binary response

For $y \in \{0,1\}$, $E(y \mid x) = P(y=1 \mid x) =: \pi(x)$. The linear model fails: fitted values leave $[0,1]$, errors are two-valued, and $\mathrm{Var}(y \mid x) = \pi(x)\{1-\pi(x)\}$ varies. $\pi(x)$ is often S-shaped, so we model

$$\pi(x) = G(\beta_0 + \beta_1 x)$$

with a cdf G (the inverse **link**): logistic $G(t) = e^t/(1+e^t)$, probit Φ, or Gumbel $\exp(-e^{-t})$. The logistic gives the **logistic regression model**

$$\pi(x) = \frac{\exp(\beta_0+\beta_1 x)}{1+\exp(\beta_0+\beta_1 x)}, \qquad \text{equivalently} \qquad \operatorname{logit}\pi(x) = \log\frac{\pi(x)}{1-\pi(x)} = \beta_0 + \beta_1 x.$$

The curve increases in x if $\beta_1 > 0$ and decreases if $\beta_1 < 0$.

Odds and odds ratios. The **odds** of $y=1$ are $\pi/(1-\pi) = P(y=1)/P(y=0) \in [0,\infty)$, with $\pi = \text{odds}/(1+\text{odds})$. Under the logistic model

$$\frac{\pi(x)}{1-\pi(x)} = e^{\beta_0}(e^{\beta_1})^x, \qquad \frac{\pi(x+1)/\{1-\pi(x+1)\}}{\pi(x)/\{1-\pi(x)\}} = e^{\beta_1}.$$

So e^{β_1} is the **odds ratio** per unit of x, the same at every x; for binary x it compares the groups, e.g. $\hat\beta_1 = 2.0$ means $e^2 \approx 7.4$ times higher odds when exposed.

16.2 Estimation

Motivation. Logistic regression models the log-odds linearly, fitted by maximum likelihood via IRLS.

With $\mathbf{x}_i = (1, x_{i1}, \ldots, x_{ip})'$ and $\pi_i = \pi(\mathbf{x}_i) = e^{\mathbf{x}_i'\beta}/(1+e^{\mathbf{x}_i'\beta})$, independent observations $y_i \sim \text{Bernoulli}(\pi_i)$ give the likelihood $L(\beta) = \prod \pi_i^{y_i}(1-\pi_i)^{1-y_i}$ and

$$l(\beta) = \sum_{i=1}^n \{y_i \log\pi_i + (1-y_i)\log(1-\pi_i)\} = \sum_{i=1}^n \{y_i \mathbf{x}_i'\beta - \log(1+e^{\mathbf{x}_i'\beta})\}.$$

Theorem 16.1

With $\pi(\beta) = (\pi_1, \ldots, \pi_n)'$ and $W(\beta) = \mathrm{diag}\{\pi_i(1-\pi_i)\}$,

$$\frac{\partial l}{\partial \beta} = X'\{\mathbf{y} - \pi(\beta)\}, \qquad \frac{\partial^2 l}{\partial\beta\partial\beta'} = -X'W(\beta)X.$$

The Hessian is negative definite (full column rank), so the MLE, if it exists, is the unique root of $X'\{\mathbf{y} - \pi(\hat\beta)\} = \mathbf{0}$.

Proof. $\frac{d}{dt}\log(1+e^t) = \frac{e^t}{1+e^t}$ and $\frac{d^2}{dt^2}\log(1+e^t) = \frac{e^t}{(1+e^t)^2} = G(t)\{1-G(t)\}$; apply the chain rule with $t_i = \mathbf{x}_i'\beta$. Since $0 < \pi_i < 1$, $\mathbf{a}'X'WX\mathbf{a} = \sum_i \pi_i(1-\pi_i)(\mathbf{x}_i'\mathbf{a})^2 > 0$ for $\mathbf{a} \neq \mathbf{0}$. ■

The likelihood equations are nonlinear and are solved by Newton–Raphson:

$$\beta^{(t+1)} = \beta^{(t)} + \{X'W^{(t)}X\}^{-1}X'\{\mathbf{y} - \pi^{(t)}\} = \{X'W^{(t)}X\}^{-1}X'W^{(t)}\mathbf{z}^{(t)},$$

with the **working response**

$$\mathbf{z}^{(t)} = X\beta^{(t)} + \{W^{(t)}\}^{-1}\{\mathbf{y} - \pi^{(t)}\}.$$

Each step is a weighted least squares regression of $\mathbf{z}^{(t)}$ on X with weights $\pi_i^{(t)}(1-\pi_i^{(t)})$:

$$\beta^{(t+1)} = \arg\min_{\beta} \{\mathbf{z}^{(t)} - X\beta\}' W^{(t)} \{\mathbf{z}^{(t)} - X\beta\},$$

hence **iteratively reweighted least squares** (IRLS), usually from $\beta^{(0)} = \mathbf{0}$. The Hessian is free of $\mathbf{y}$, so Newton–Raphson is Fisher scoring, and $\{X'W(\hat\beta)X\}^{-1}$ estimates $\operatorname{Cov}\hat\beta$. Wald: $\hat\beta_j/\operatorname{se}(\hat\beta_j) \approx N(0,1)$; LR: $2\{l(\hat\beta_F) - l(\hat\beta_R)\}$ vs chi-square.

If the two classes can be perfectly separated by a hyperplane in x, l increases without bound along a direction and the MLE does not exist.

16.3 Two-by-two tables

Motivation. With one binary predictor the model is a 2×2 table and the slope a log odds ratio.

Binary exposure (x) and outcome (y) data arise from three designs:

- **cohort** (prospective) studies follow exposed and unexposed groups and record outcomes;
- **case–control** (retrospective) studies sample cases ($y = 1$) and controls ($y = 0$) and record past exposure;
- **cross-sectional** studies sample individuals and record both at one time.

With counts n_{xy}:

	$y=0$	$y=1$	total
$x=0$	n_{00}	n_{01}	n_{0+}
$x=1$	n_{10}	n_{11}	n_{1+}

The **odds ratio** is $OR = \dfrac{\pi_1/(1-\pi_1)}{\pi_0/(1-\pi_0)}$ with $\pi_x = P(y=1 \mid x)$, estimated by $\widehat{OR} = \dfrac{n_{11}n_{00}}{n_{10}n_{01}}$.

Theorem 16.2

In the logistic model $\operatorname{logit} P(y=1 \mid x) = \beta_0 + \beta_1 x$ with binary x, the MLEs satisfy

$$\hat\pi_1 = \frac{e^{\hat\beta_0+\hat\beta_1}}{1+e^{\hat\beta_0+\hat\beta_1}} = \frac{n_{11}}{n_{1+}}, \qquad \hat\pi_0 = \frac{e^{\hat\beta_0}}{1+e^{\hat\beta_0}} = \frac{n_{01}}{n_{0+}}, \qquad e^{\hat\beta_1} = \frac{n_{11}n_{00}}{n_{10}n_{01}}.$$

Proof. The log-likelihood is

$$l = n_{11}(\beta_0+\beta_1) + n_{01}\beta_0 - n_{1+}\log(1+e^{\beta_0+\beta_1}) - n_{0+}\log(1+e^{\beta_0}).$$

The model is saturated (two parameters for two probabilities), so by invariance the MLEs of π_1, π_0 are the sample proportions n_{11}/n_{1+} and n_{01}/n_{0+}. Then $e^{\hat\beta_1} = \frac{\hat\pi_1/(1-\hat\pi_1)}{\hat\pi_0/(1-\hat\pi_0)} = \frac{n_{11}/n_{10}}{n_{01}/n_{00}}$. ∎

Why the odds ratio. In case–control studies row proportions do not estimate π_x (case and control numbers are fixed), but the odds ratio is symmetric in x and y:

$$\frac{P(y=1 \mid x=1)P(y=0 \mid x=0)}{P(y=0 \mid x=1)P(y=1 \mid x=0)} = \frac{P(x=1 \mid y=1)P(x=0 \mid y=0)}{P(x=0 \mid y=1)P(x=1 \mid y=0)},$$

both being $\frac{p_{11}p_{00}}{p_{10}p_{01}}$; so β_1 is estimable from case–control data, only the intercept being distorted.

Example 16.1

140 subjects get a supplement, 140 a placebo; 17 and 31 catch a cold. With $x = 1$ for the supplement, $\widehat{OR} = \frac{17 \cdot 109}{123 \cdot 31} = 0.49$, $\hat{\beta}_1 = -0.72$, SE $\sqrt{\frac{1}{17} + \frac{1}{123} + \frac{1}{31} + \frac{1}{109}} = 0.33$, Wald -2.19: significant at 5%.

16.4 Selected exercises

Motivation. Fitted and observed counts agree, and the SE of Example 16.1 comes from the information matrix.

Exercise 16.1 (Fitted counts equal observed counts)

(a) Show that the MLE of a logistic regression satisfies $X'\hat{\pi} = X'\mathbf{y}$; in particular, with an intercept, $\sum_i \hat{\pi}_i = \sum_i y_i$.
(b) For a binary x with all four cell counts n_{xy} positive, derive Theorem 16.2 directly from (a).

Solution [Supplement].

(a) The MLE solves $X'\{\mathbf{y} - \pi(\hat{\beta})\} = \mathbf{0}$ (Theorem 16.1); the intercept row gives $\sum \hat{\pi}_i = \sum y_i$, and each regressor $\sum_i x_{ij}\hat{\pi}_i = \sum_i x_{ij} y_i$.

(b) With $\mathbf{x}_i = (1, x_i)'$ and $x_i \in \{0, 1\}$, $\hat{\pi}_i$ takes the value $\hat{\pi}_1$ when $x_i = 1$ and $\hat{\pi}_0$ when $x_i = 0$. The two equations of (a) read

$$n_{0+}\hat{\pi}_0 + n_{1+}\hat{\pi}_1 = n_{01} + n_{11}, \qquad n_{1+}\hat{\pi}_1 = n_{11}.$$

So $\hat{\pi}_1 = n_{11}/n_{1+}$, $\hat{\pi}_0 = n_{01}/n_{0+}$ (in $(0, 1)$), and $\hat{\beta}_0 = \operatorname{logit} \hat{\pi}_0$, $\hat{\beta}_1 = \operatorname{logit} \hat{\pi}_1 - \operatorname{logit} \hat{\pi}_0$ are the unique MLE (strict concavity), with $e^{\hat{\beta}_1} = \frac{n_{11}/n_{10}}{n_{01}/n_{00}}$. ■

Exercise 16.2 (Standard error of the log odds ratio)

In the setting of Exercise 16.1(b), show that the entry of $\{X'W(\hat{\beta})X\}^{-1}$ for β_1 equals

$$\frac{1}{n_{11}} + \frac{1}{n_{10}} + \frac{1}{n_{01}} + \frac{1}{n_{00}},$$

the squared standard error used in Example 16.1.

Solution [Supplement].

Step 1. $X'WX = \sum_i w_i \mathbf{x}_i \mathbf{x}_i'$ with $w_i = \hat{\pi}_i(1 - \hat{\pi}_i)$. Grouping by x, with $a = n_{0+}\hat{\pi}_0(1 - \hat{\pi}_0)$ and $b = n_{1+}\hat{\pi}_1(1 - \hat{\pi}_1)$,

$$X'WX = a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} a + b & b \\ b & b \end{pmatrix}.$$

Step 2. Its determinant is $(a+b)b - b^2 = ab > 0$, and the $(2, 2)$ entry of the inverse is $(a+b)/(ab) = 1/a + 1/b$.

Step 3. By Exercise 16.1(b), $a = n_{0+} \cdot \frac{n_{01}}{n_{0+}} \cdot \frac{n_{00}}{n_{0+}} = \frac{n_{01} n_{00}}{n_{0+}}$, so $\frac{1}{a} = \frac{n_{00} + n_{01}}{n_{01} n_{00}} = \frac{1}{n_{01}} + \frac{1}{n_{00}}$. In the same way $\frac{1}{b} = \frac{1}{n_{11}} + \frac{1}{n_{10}}$. Adding gives the claim. ■